When and How Often is Weighted Majority Vote Optimal Under Log Loss?

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Abstract

Suppose one has labeling functions (LFs, e.g. "document contains 'birdie' ⇒ 'sports' is its topic") that predict on unlabeled datapoints. How should one combine their predictions to get a single labeling when LFs have different accuracies and are sometimes contradictory? One strategy is to do a weighted majority vote over their predictions. We study the optimality of weighting strategies under log loss, i.e. functions that take LF accuracies, predictions, etc., and return LF weights.

Formal Model

- n unlabeled datapoints
$$X = \{x_1, x_2, \dots, x_n\} \subset \mathcal{X}$$
- k labels denoted $\mathcal{Y} = \{1, 2, \dots, k\}$
- p labeling functions (LFs):
$$h^{(1)}, h^{(2)}, \dots, h^{(p)}\colon X \to \mathcal{Y}$$

Goal: For each datapoint x_i and labels $\ell \in \{1, 2, \dots, k\}$, infer the underlying conditional label probability
$$\eta_{i\ell} = \Pr(y_i = \ell \mid x_i).$$

- We'll estimate this using (a function of) a weighted majority vote over the LF predictions.

A First Pass Approach: Majority Vote (MV)

Each LF receives weight 1. This does well when the majority of LFs predict the correct label, and is often good in practice. However, it is not hard to create situations where MV does poorly.

Some Failure Modes

- Large class imbalance along with LFs that have low accuracies
  - Here, it's often better to predict the majority class and ignore LFs altogether
- Large number of spammers (LFs with accuracy $\approx 1/k$)
  - If there are only a few high accuracy LFs, their votes get drowned out by the spammers' votes (as each spammer gets weight 1)

An Attempted Fix: One-Coin Dawid-Skene (OCDS)

To try and fix MV's modes of failures one might assign weights to LFs more cleverly. For this strategy, LF $h^{(j)}$ gets weight θ_j proportional to its accuracy and class ℓ gets weight ω_ℓ proportional to its frequency.

- If $b_j^* = Pr(h^{(j)}(x) = y)$, $w_\ell^* = Pr(y = \ell)$:
$$\theta_j = \log\left(\frac{b_j^*(k-1)}{1 - b_j^*}\right), \quad \omega_\ell = \log w_\ell^*$$
- The OCDS estimate of $\eta_{i\ell}$ is
$$Pr(y = \ell \mid h^{(j)}(x_i), \forall j \in [p]) \propto \exp\left(\omega_\ell + \sum_{j=1}^{p} \theta_j \mathbf{1}(h^{(j)}(x_i) = \ell)\right)$$

Are MV's modes of failure addressed?

- Large class imbalance along with LFs that have low accuracies
  - For any minority class ℓ s.t. $w_\ell^* \ll 1$, it gets weight $\omega_\ell \ll 0$
- Large number of spammers (LFs with accuracy $\approx 1/k$)
  - $\theta_j \approx 0$ whenever $b_j^* \approx \frac{1}{k}$ so spammers get little to no weight

Problem: For almost all X, these θ_j, ω_ℓ are not optimal with respect to log loss!

Simple and Complex Weighting Strategies

Weighting strategies are functions convert information such as LF accuracies, class frequencies, etc., into weights.

Simple Weighting Strategies

These strategies *only* use LF accuracies and class frequencies:
$$\theta\colon [0,1]^p \times \Delta_k \to \mathbb{R}^p, \quad \omega\colon [0,1]^p \times \Delta_k \to \mathbb{R}^k$$

- Majority Vote: $\theta = \mathbf{1}_p, \omega = \mathbf{0}_k$
- One-Coin Dawid-Skene:
$$\theta = \log\left(\frac{b(k-1)}{1-b}\right), \quad \omega = \log(w)$$

Complex Weighting Strategies

These weighting strategies use LF accuracies, class frequencies, *and* the LF predictions. This difference is crucial!

Our Contributions

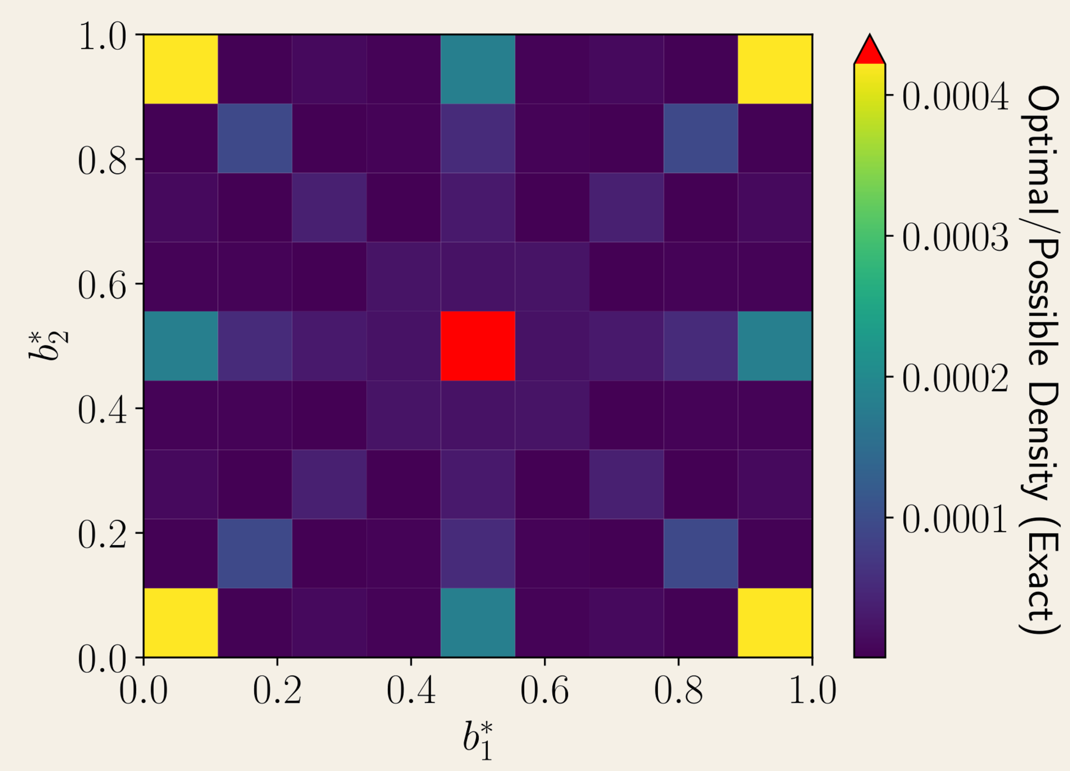
Simple Weighting Strategies are Almost Never Optimal

Fix LF accuracies b^*, class frequencies w^*, and consider an arbitrary dataset X. If we use a simple weighting strategy to estimate η (for that X), then for almost every X, our estimate is not optimal with respect to log loss.

Why? Too many bad cases. E.g. consider the case of duplicated LFs. Simple weighting strategies cannot take advantage of this info – duplicated LFs will have the same accuracies, hence the duplicated prediction receives weight proportional to the number of duplicates.

How do we avoid this? Use complex weighting strategies, e.g. Balsubramani-Freund, if one requires optimality.

Experimental Result: For $p = k = 2$, $w^* = \mathbf{1}_2/2$, we compute the exact fraction of datasets where OCDS is optimal.



Exact Analysis of Error from Weighted Majority Vote

Consider two arbitrary weighting strategies (θ, ω) and (θ', ω').

Weighting Strategy Comparison: An exact expression for the extra error incurred by (θ, ω) over (θ', ω') is given. We also give a test to determine whether the extra error is > 0.

Covariate Shift Analysis: Suppose (θ, ω) is better than (θ', ω'). We exactly characterize how much covariate shift can be tolerated before (θ', ω') becomes better than (θ, ω).